| DS | **Session** | Linked List | **Question Information** | Level 1 Challenge 34 |
|---|---|---|---|---|

**Question description**

Problem Description:

For Engineering Second -year students, the instructor is organising a surprise test.

He's given you a new node is added before the given node of the given Linked List.

For example if the given Linked List is 5->10->15->20->25 and

we add an item 30 before node 15, then the Linked List becomes 5->10->30->15->20->25.

Since a Linked List is typically represented by the head of it, we have to traverse the list till node and then insert the node.

**Constraints:**

1 < arr <100

**Input Format**

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third line contains the node P before which the node to be inserted.

Fourth line contain the node X to be inserted.

**Output Format**

case 1(node P found in Linked list) :

Display the final Linked List.

case 2(node P not found in Linked list) :

Print Node not found!

Display the Linked list.

## Logical Test Cases

### Test Case 1

INPUT (STDIN)

```
4
115 210 116 110
116
17
```

EXPECTED OUTPUT

```
Linked List:->115->210->17->116->110
```

### Test Case 2

INPUT (STDIN)

```
4
15 20 16 10
25
17
```

EXPECTED OUTPUT

```
Node not found!
Linked List:->15->20->16->10
```

## Mandatory Test Cases

### Test Case 1

KEYWORD

```
struct node
```

### Test Case 2

KEYWORD

```
node *next;
```

### Test Case 3

KEYWORD

```
*start
```

### Test Case 4

KEYWORD

```
p2->next=p1;
```

### Test Case 5

KEYWORD

```
void display()
```

Complexity Test Cases

| Test Case 1 | Test Case 2 | Test Case 3 |
| --- | --- | --- |
| CYCLOMATIC COMPLEXITY | TOKEN COUNT | NLOC |
| 3 | 275 | 66 |

# Code

solution.c

```c
#include <stdio.h>
#include <stdlib.h>

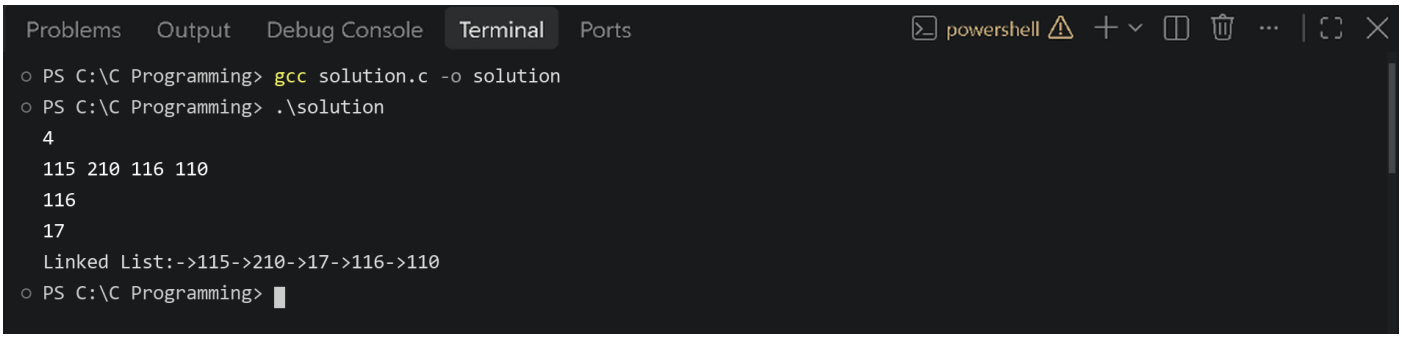
struct node {
    int data;
    struct node *next;
};

struct node *start = NULL;

void display() {}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    struct node *tail = NULL;
    for (int i = 0; i < n; i++) {
        int v; scanf("%d", &v);
        struct node *t = (struct node*)malloc(sizeof(struct node));
        t->data = v; t->next = NULL;
        if (!start) start = tail = t;
        else { tail->next = t; tail = t; }
    }
    int p, x;
    scanf("%d %d", &p, &x);
    struct node *cur = start, *prev = NULL;
    int found = 0;
    while (cur) {
        if (cur->data == p) { found = 1; break; }
        prev = cur;
        cur = cur->next;
    }
    if (!found) {
        printf("Node not found!\nLinked List:");
        cur = start;
        while (cur) { printf("->%d", cur->data); cur = cur->next; }
        printf("\n");
    } else {
        struct node *newNode = (struct node*)malloc(sizeof(struct node));
        newNode->data = x;
        newNode->next = cur;
        if (prev == NULL) start = newNode;
        else prev->next = newNode;
        printf("Linked List:");
        cur = start;
        while (cur) { printf("->%d", cur->data); cur = cur->next; }
        printf("\n");
    }
    return 0;
}
```

## Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
4
115 210 116 110
116
17
Linked List:->115->210->17->116->110
PS C:\C Programming>